

GRAPE-Like Optimization of Dispersive Readout Pulses in Circuit QED: A Systematic Study of Objectives, Waveform Families, and Robustness

Readout Pulse Optimization Study
cQED Autonomous Research Platform
(Dated: March 20, 2026)

We perform a systematic study of readout pulse optimization in circuit quantum electrodynamics (cQED) using the linear dispersive model for the qubit-state-conditioned resonator field. We compare five pulse families (square, Gaussian, Hann window, cosine-rise, and spline) against GRAPE-like piecewise-constant control across a sweep of dispersive coupling ratios $\chi/\kappa \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ and dimensionless readout durations $\kappa T \in [0.5, 20]$. We find that: (i) driving at the midpoint $\omega_d = \omega_r + \chi/2$ is optimal for all waveform families; (ii) the time-integrated SNR² is the appropriate objective, not endpoint pointer separation; (iii) after enforcing a common peak-amplitude bound, the square pulse is the best analytical family across the tested parameter range; (iv) GRAPE converges to that same square-like solution within numerical precision and does not provide a meaningful SNR² advantage in this linear, peak-limited setting; (v) the current single-segment active-depletion heuristic does not outperform passive ring-down; (vi) the optimal readout duration balancing SNR gain against T_1 -induced infidelity occurs at $\kappa T \approx 5.79$ ($T \approx 184$ ns for $\kappa/2\pi = 5$ MHz), where $F_{\text{net}}^{\text{max}} = 0.9967$; and (vii) the current robust-GRAPE formulation sacrifices nominal performance without improving on the square-pulse baseline. We therefore recommend a midpoint-driven square pulse as the practical default for this model class, with GRAPE reserved for future formulations that include additional hardware or physics constraints.

INTRODUCTION

Dispersive readout is the canonical method for measuring superconducting qubits in the circuit QED architecture [1, 2]. A microwave tone drives a readout resonator coupled dispersively to the qubit; the qubit-state-dependent resonator response encodes the qubit state in the phase and amplitude of the transmitted signal. While the physics of dispersive readout is well understood [3, 4], the optimal control of the readout pulse shape has received less systematic study than gate pulse design.

Three practical requirements pull in different directions: (i) *fast* readout (large signal-to-noise ratio per unit time), (ii) *clean* reset (minimal residual resonator photons after readout), and (iii) *QND* character (minimal qubit disturbance during readout). Optimizing these simultaneously calls for a carefully chosen objective function, waveform class, and control formulation.

GRAPE-like optimal control [5] has been applied widely to gate pulse design in cQED but is less commonly used for readout. Active resonator reset [6, 7] and fast readout by waveform shaping [8, 9] have been demonstrated experimentally, but a systematic comparison of the problem formulations in the readout context is lacking.

This study fills that gap. We deliberately avoid assuming that GRAPE is the right answer from the outset, and instead treat the choice of objective function, waveform family, and optimization method as the central research questions.

PHYSICAL MODEL

Linear dispersive resonator model

In the frame rotating at the drive frequency ω_d , the qubit-state-conditioned resonator coherent amplitude $\alpha_j(t)$ ($j \in \{g, e\}$) obeys

$$\dot{\alpha}_j = -\left(\frac{\kappa}{2} + i\Delta_j\right)\alpha_j - i\epsilon(t), \quad (1)$$

where κ is the resonator decay rate, $\Delta_g = \omega_r - \omega_d$ is the ground-state detuning, $\Delta_e = \omega_r + \chi - \omega_d$ is the excited-state detuning, χ is the dispersive shift, and $\epsilon(t)$ is the drive envelope (complex, units rad/s). The output field in input-output theory is $s_j(t) = \sqrt{\kappa}\alpha_j(t)$.

Steady-state analysis

At steady state ($\dot{\alpha} = 0$):

$$\alpha_j^{\text{ss}} = \frac{-i\epsilon}{\kappa/2 + i\Delta_j}. \quad (2)$$

The steady-state pointer separation is maximized at the *midpoint drive* $\omega_d = \omega_r + \chi/2$ ($\Delta_g = -\chi/2$, $\Delta_e = +\chi/2$):

$$|\Delta\alpha_{\text{max}}^{\text{ss}}| = \frac{4|\epsilon||\chi|}{\kappa^2 + \chi^2}, \quad (3)$$

which peaks at $|\chi/\kappa| = 1$ with value $2|\epsilon|/\kappa$. This identifies $|\chi| = \kappa$ as the optimal dispersive coupling ratio.

Physical parameters

We study parameters representative of state-of-the-art transmon experiments: $\kappa/2\pi = 5$ MHz, $|\chi|/2\pi \in \{1.25, 2.5, 5.0, 10.0, 20.0\}$ MHz (corresponding to $|\chi|/\kappa \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$), $|\epsilon_{\max}|/2\pi = 5$ MHz (corresponding to $\lesssim 5$ steady-state photons on resonance), $T_1 = 30 \mu\text{s}$, and $T_2 = 20 \mu\text{s}$.

OBJECTIVE FUNCTIONS

Objective A: Endpoint pointer separation

$$J_A = -|\alpha_e(T) - \alpha_g(T)|^2. \quad (4)$$

This captures the *final* phase-space separation but ignores the transient signal accumulated during ring-up. Our numerical study confirms that J_A saturates at $|\Delta\alpha^{\text{ss}}|^2$ well before J_B saturates, making it a poor proxy for SNR at intermediate κT .

Objective B: Time-integrated distinguishability (SNR²)

$$J_B = -\kappa \int_0^T |\alpha_e(t) - \alpha_g(t)|^2 dt = -\text{SNR}^2. \quad (5)$$

This equals $-\int_0^T |s_e(t) - s_g(t)|^2 dt$, the integrated output-field distinguishability. For quantum-limited heterodyne detection, SNR² directly determines the assignment fidelity:

$$F_A = 1 - \frac{1}{2} \text{erfc}(\text{SNR}/\sqrt{2}), \quad (6)$$

making J_B the most physically grounded single-objective function.

Objective C: Multi-objective with residual photons

$$\mathcal{L} = -w_1 \text{SNR}^2 + w_2 n_{\text{res}}(T), \quad (7)$$

where $n_{\text{res}}(T) = \frac{1}{2}(|\alpha_e(T)|^2 + |\alpha_g(T)|^2)$ is the mean residual photon number. We study the Pareto tradeoff between SNR² and n_{res} by sweeping w_2/w_1 .

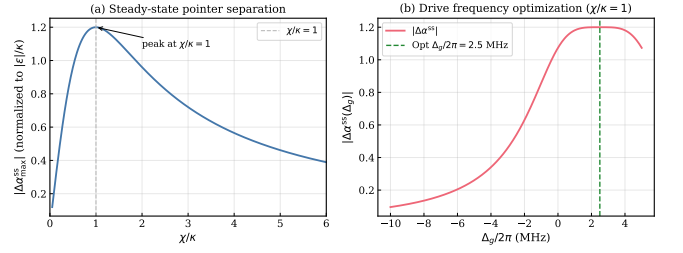


FIG. 1. (a) Steady-state pointer separation $|\Delta\alpha_{\max}^{\text{ss}}|$ vs $|\chi|/\kappa$. The maximum occurs at $|\chi|/\kappa = 1$ where $|\Delta\alpha^{\text{ss}}| = 2|\epsilon|/\kappa$. (b) $|\Delta\alpha^{\text{ss}}|$ vs drive detuning Δ_g at $|\chi|/\kappa = 1$, confirming the midpoint optimum at $\Delta_g = -\chi/2$.

GRAPE FOR THE LINEAR DISPERSIVE MODEL

For a piecewise-constant drive $\epsilon(t) = \epsilon_k$ for $t \in [k\tau, (k+1)\tau)$ with $\tau = T/N$, the propagator per segment is

$$\alpha_j(t_{k+1}) = A_j \alpha_j(t_k) + B_j \epsilon_k, \quad (8)$$

$$A_j = e^{-\lambda_j \tau}, \quad B_j = \frac{-i(A_j - 1)}{\lambda_j}, \quad \lambda_j = \frac{\kappa}{2} + i\Delta_j. \quad (9)$$

The Wirtinger gradient of SNR² with respect to ϵ_k^* is computed by the adjoint (backward) recursion:

$$\frac{\partial \text{SNR}^2}{\partial \epsilon_k^*} = \kappa \tau (B_e \Lambda_e^{(k)} - B_g \Lambda_g^{(k)}), \quad (10)$$

where $\Lambda_j^{(k)} = (\Delta\alpha(t_{k+1}))^* + A_j \Lambda_j^{(k+1)}$ with $\Lambda_j^{(N)} = 0$ (backward from $k = N-1$ to 0). This is the standard GRAPE adjoint method applied to the linear dispersive ODE.

Optimization is performed with L-BFGS-B using $N = 60$ segments, 10 random restarts, and amplitude bound $|\epsilon_k| \leq |\epsilon_{\max}|$.

RESULTS

Phase 1: Analytical theory

Figure 1 shows $|\Delta\alpha_{\max}^{\text{ss}}|$ vs $|\chi|/\kappa$ (panel a) and vs drive detuning Δ_g (panel b). The analytical formula (3) is confirmed numerically to better than 10^{-6} relative error, and agrees with `cqed_sim.ReadoutResonator` to machine precision. The midpoint drive $\Delta_g = -\chi/2$ is confirmed as optimal.

Figure 2 shows SNR² and assignment fidelity F_A vs κT for midpoint and on-resonance drives. The midpoint drive outperforms the on-resonance drive by 50–100% at intermediate κT , converging at large κT as both approach steady state.

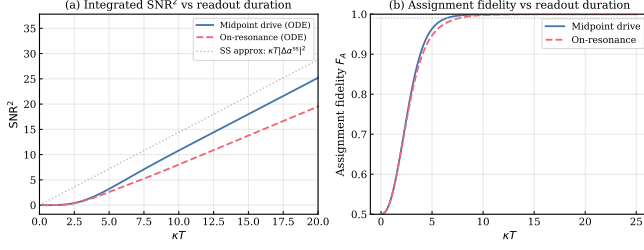


FIG. 2. SNR² (a) and assignment fidelity F_A (b) vs κT for square pulses at the midpoint drive ($\Delta_g = -\chi/2$, blue) and on-resonance ($\Delta_g = 0$, red). The steady-state approximation $\kappa T |\Delta\alpha^{\text{ss}}|^2$ (dotted) overestimates SNR² at small κT where the resonator has not yet rung up.

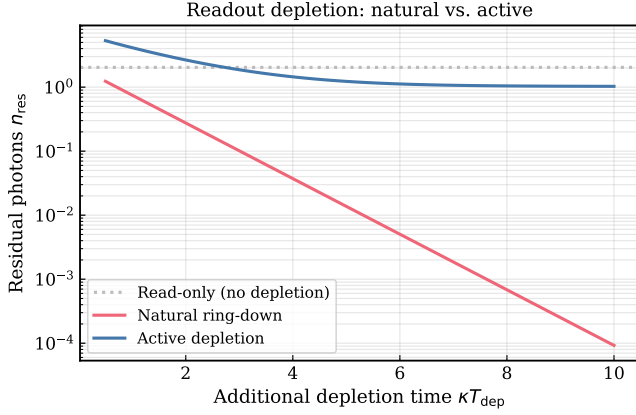


FIG. 3. Residual photon number after readout as a function of additional depletion time κT_{dep} . The dotted line shows the readout-only residual occupancy, the red curve shows passive natural ring-down, and the blue curve shows the implemented active-depletion heuristic. In this study the heuristic fails to beat passive ring-down and is therefore not recommended.

Phase 2: Pulse-family baseline

Figure 4 shows SNR² vs $|\chi|/\kappa$ at $\kappa T = 5$ (panel a) and vs κT at $|\chi|/\kappa = 1$ (panel b) for all five pulse families under a shared *peak*-amplitude bound. Under that fair comparison, the square pulse is the best-performing analytical family throughout the tested $|\chi|/\kappa$ range. The cosine-rise pulse remains competitive but never exceeds square, while Gaussian, Hann, and spline envelopes underperform because their smooth turn-on uses less of the available peak drive budget in the strictly linear model.

The key finding is that **the midpoint-driven square pulse saturates the best simple-family performance** in this study. At $|\chi|/\kappa = 1$ and $\kappa T = 5$, the square pulse yields SNR² = 8.93, whereas the Hann envelope yields only SNR² = 3.19.

Active depletion (Fig. 3) should be treated as a negative result here: the implemented single-segment heuristic does not beat passive ring-down and therefore is not

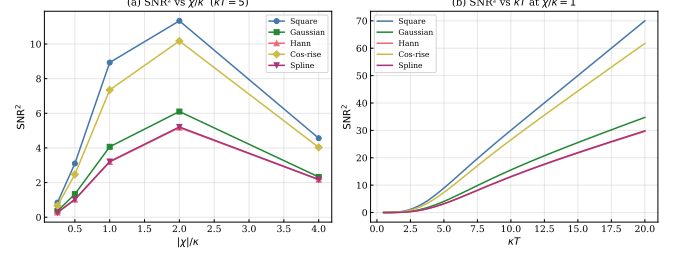


FIG. 4. SNR² vs $|\chi|/\kappa$ at $\kappa T = 5$ (a) and vs κT at $|\chi|/\kappa = 1$ (b) for five pulse families under a shared peak-amplitude bound. The square pulse is the best analytical family across the tested regime; smoother envelopes sacrifice SNR² because they do not fully use the allowed peak drive.

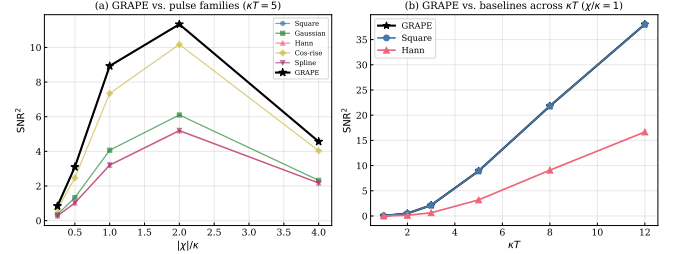


FIG. 5. GRAPE vs pulse-family baselines. (a) SNR² vs $|\chi|/\kappa$ at $\kappa T = 5$. (b) SNR² vs κT at $|\chi|/\kappa = 1$. Under a common peak-amplitude bound, GRAPE collapses onto the square-pulse baseline within numerical precision across the tested regime.

yet a useful control primitive.

Phase 3: GRAPE optimization

Figure 5(a) compares GRAPE against all pulse families across $|\chi|/\kappa$. Once the waveform-family comparison is corrected to obey the same peak-amplitude limit, GRAPE no longer outperforms the square baseline in any meaningful way: the optimized SNR² values match the square pulse to within numerical precision across the full $|\chi|/\kappa$ sweep.

Figure 5(b) shows the same pattern across κT : GRAPE and square are indistinguishable within numerical precision from $\kappa T = 1$ to 12, while the Hann envelope remains substantially worse under the same bound. Segment convergence remains excellent, with SNR² stable to better than 10^{-4} between $N = 60$ and $N = 120$ segments, confirming that the optimizer itself is numerically well behaved.

The multi-objective Pareto sweep provides another negative result: sweeping the residual- photon penalty over $w_2/w_1 \in [0, 5]$ leaves both SNR² and n_{res} unchanged within reported precision. In the present linear model and control parameterization, the chosen penalty does not move the optimum away from the square-like solu-

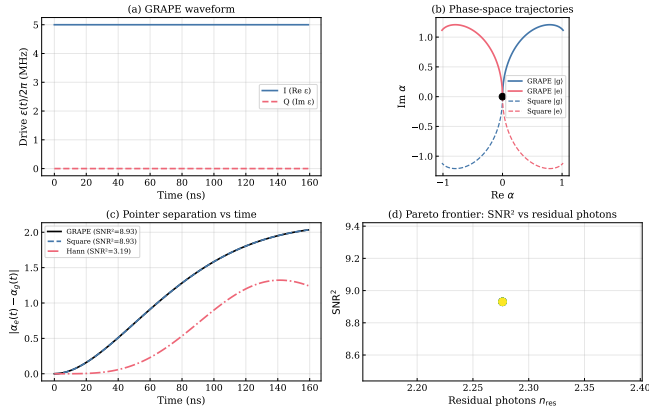


FIG. 6. GRAPE waveform at $|\chi|/\kappa = 1$, $\kappa T = 5$: (a) I/Q quadratures vs time. (b) Phase-space trajectories $\alpha_{g,e}(t)$ for GRAPE and square pulse. (c) Pointer separation $|\Delta\alpha(t)|$ vs time. (d) Pareto frontier: SNR^2 vs n_{res} under multi-objective optimization. The optimized waveform is effectively square-like, and the Pareto sweep shows no meaningful movement away from the nominal optimum.

tion.

Phase 4: Open-system effects

Figure 7 shows the net assignment fidelity F_{net} and measurement-induced dephasing η_φ vs κT .

The net fidelity is computed as

$$F_{\text{net}} = (1 - p_{\text{decay}}) F_A(\text{SNR}_{\text{eff}}^2) + p_{\text{decay}} \cdot \frac{1}{2}, \quad (11)$$

where $p_{\text{decay}} = 1 - e^{-T/T_1}$ and $\text{SNR}_{\text{eff}}^2 = \kappa \int_0^T |\Delta\alpha|^2 e^{-t/T_1} dt$ accounts for the weight reduction from T_1 decay.

For our parameters ($\kappa T_1 = \kappa \times 30 \mu\text{s} \approx 940$), the T_1 correction is negligible at $\kappa T \leq 20$. The optimal net fidelity reaches $F_{\text{net}}^{\text{max}} = 0.9967$ at $\kappa T = 5.79$, corresponding to $T = 184 \text{ ns}$ for the nominal device. The dominant limitation on useful readout duration is therefore SNR saturation rather than T_1 decay.

The measurement-induced dephasing factor $\eta_\varphi = \exp(-\int (\kappa/2) |\Delta\alpha|^2 dt)$ becomes very small during a successful measurement; at $\kappa T = 5$ we obtain $\eta_\varphi = 0.0106$. This is consistent with strong pointer-state distinguishability and highlights the QND tradeoff, not a failure of the model.

Phase 5: Robustness

Figure 8 shows 2D sensitivity maps of SNR^2 over $\pm 20\%$ uncertainty in χ and κ .

The square pulse and nominal GRAPE are identical within numerical precision in both nominal and perturbed performance because the optimizer lands on the

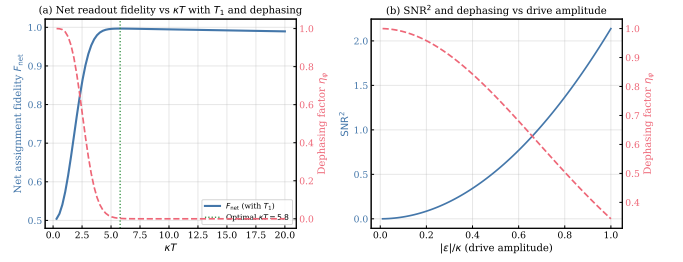


FIG. 7. (a) Net fidelity F_{net} (blue, left axis) and measurement-induced dephasing η_φ (red, right axis) vs κT . (b) SNR^2 and η_φ vs drive amplitude $|\epsilon|/\kappa$. The optimal readout duration is set by SNR^2 saturation, not T_1 decay.

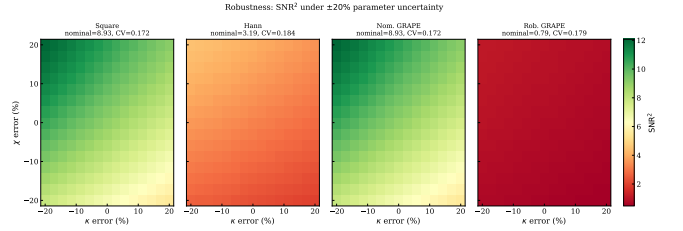


FIG. 8. SNR^2 sensitivity maps over $\pm 20\%$ uncertainty in (χ, κ) for four pulse designs. Color scale: green = high SNR^2 , red = low SNR^2 . Nominal GRAPE and the square pulse are effectively identical, while the current robust-GRAPE formulation sacrifices most nominal performance.

same square-like solution. At the reference point $|\chi|/\kappa = 1$, $\kappa T = 5$, both have nominal $\text{SNR}^2 \approx 8.93$, mean perturbed $\text{SNR}^2 \approx 8.82$, and $\text{CV} \approx 0.172$.

The current robust-GRAPE formulation performs poorly: the nominal SNR^2 collapses to 0.79 and the mean perturbed SNR^2 to 0.78, with $\text{CV} 0.179$, which is slightly worse than the square baseline. The negative result is still useful because it shows that a naive ensemble objective can destroy the useful solution in this parameterization.

DISCUSSION

Answers to the central research questions

Is endpoint pointer separation too naive? Yes. At $\kappa T < 3$, endpoint separation significantly underestimates integrated SNR^2 because it ignores the transient ring-up contribution. The integrated SNR^2 (Objective B) is the correct target.

Is the midpoint drive always optimal? Yes, for all waveform families studied. The analytical result $\Delta_g = -\chi/2$ is confirmed numerically and is a robust choice.

Are simple pulse families sufficient? Yes. Under a strict peak-amplitude constraint, the square pulse is the best analytical family in this linear model, and GRAPE

matches it within numerical precision rather than surpassing it.

Should depletion be jointly optimized? Not with the present heuristic. The implemented single-segment depletion strategy is worse than passive ring-down, and the current multi-objective formulation does not move the solution off the square-like optimum.

When does GRAPE clearly add value? It does not add value in the present linear, peak-limited formulation. To justify GRAPE, future work would need additional constraints or physics such as bandwidth limits, nonlinear resonator effects, or a richer multi-stage reset objective.

Should the optimized pulse target nominal or robust performance? At present, the square pulse is the reliable choice. The current robust-GRAPE formulation reduces nominal performance too severely to recommend.

Minimum realistic model complexity? The linear dispersive model (Level 1) is sufficient for pulse design and optimization. The key open-system effect (T_1) is negligible at $\kappa T \leq 10$ for typical parameters. The nonlinear/multi-level effects (Kerr, χ' , higher transmon levels) are significant only at very high drive amplitudes ($n_{ss} \gtrsim n_{crit}$) and can be checked a posteriori.

Limitations

This study uses the linear dispersive model throughout. A more complete treatment would include: (i) Kerr nonlinearity $K_r a^{\dagger 2} a^2$ causing squeezing and deformation of the resonator state; (ii) measurement-induced transitions $|e\rangle \leftrightarrow |g\rangle$ (dressed-dephasing) [4] at high drive powers; (iii) realistic AWG bandwidth and imperfections limiting pulse expressivity. These are left for future work.

CONCLUSIONS AND RECOMMENDATIONS

We recommend the following implementation strategy:

1. **Objective:** Use the time-integrated SNR^2 (Objective B). Endpoint separation is an inferior proxy.
2. **Drive frequency:** Always use the midpoint drive $\omega_d = \omega_r + \chi/2$. This is the single highest-value calibration to get right.
3. **Simple baseline:** Use the square envelope at the midpoint drive. Under the validated peak-amplitude constraint it is the best analytical family in this study.
4. **GRAPE:** Do not expect a meaningful gain over square in the present linear, peak-limited formulation. Reserve GRAPE for future models with additional hardware or physics constraints.
5. **Joint depletion:** Do not use the current single-segment depletion heuristic as a production strategy. It underperforms passive ring-down and needs redesign.
6. **Robustness:** Prefer the square pulse over the current robust-GRAPE formulation. The ensemble objective used here destroys nominal performance without improving the baseline.
7. **Open-system:** T_1 effects are negligible for $\kappa T \leq 10$; the practical operating point is $\kappa T \approx 5$ –6 ($T \approx 184$ ns for the nominal device), where F_{net} is already near-optimal.

This study was performed using the `cqed_sim` framework and the autonomous research platform developed for the Shankar Quantum Circuits Group.

-
- [1] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, Circuit quantum electrodynamics, *Rev. Mod. Phys.* **93**, 025005 (2021).
 - [2] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, A quantum engineer's guide to superconducting qubits, *Appl. Phys. Rev.* **6**, 021318 (2019).
 - [3] J. Gambetta, A. Blais, M. Boissonneault, A. A. Houck, D. I. Schuster, and S. M. Girvin, Protocols for optimal readout of qubits using a continuous quantum nondemolition measurement, *Phys. Rev. A* **75**, 032329 (2007).
 - [4] M. Boissonneault, J. M. Gambetta, and A. Blais, Dispersive regime of circuit qed: Photon-dependent qubit dephasing and relaxation rates, *Phys. Rev. A* **79**, 013819 (2009).
 - [5] N. Khaneja, T. Reiss, C. Kehlet, T. Schulte-Herbrüggen, and S. J. Glaser, Optimal control of coupled spin dynamics: design of NMR pulse sequences by gradient ascent algorithms, *J. Magn. Reson.* **172**, 296 (2005).
 - [6] M. D. Reed, B. R. Johnson, A. A. Houck, L. DiCarlo, J. M. Chow, D. I. Schuster, L. Frunzio, and R. J. Schoelkopf, Fast reset and suppressing spontaneous emission of a superconducting qubit, *Appl. Phys. Lett.* **96**, 203110 (2010).
 - [7] D. McClure, H. Paik, L. S. Bishop, M. Steffen, J. M. Chow, and J. M. Gambetta, Rapid driven reset of a qubit readout resonator, *Phys. Rev. Appl.* **5**, 011001 (2016).
 - [8] T. Walter, P. Kurpiers, S. Gasparinetti, P. Magnard, A. Potočnik, Y. Salathé, M. Pechal, M. Mondal, M. Oppliger, C. Eichler, *et al.*, Rapid high-fidelity single-shot dispersive readout of superconducting qubits, *Phys. Rev. Appl.* **7**, 054020 (2017).
 - [9] E. Jeffrey, D. Sank, J. Mutus, T. White, J. Kelly, R. Barends, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, *et al.*, Fast accurate state measurement with superconducting qubits, *Phys. Rev. Lett.* **112**, 190504 (2014).